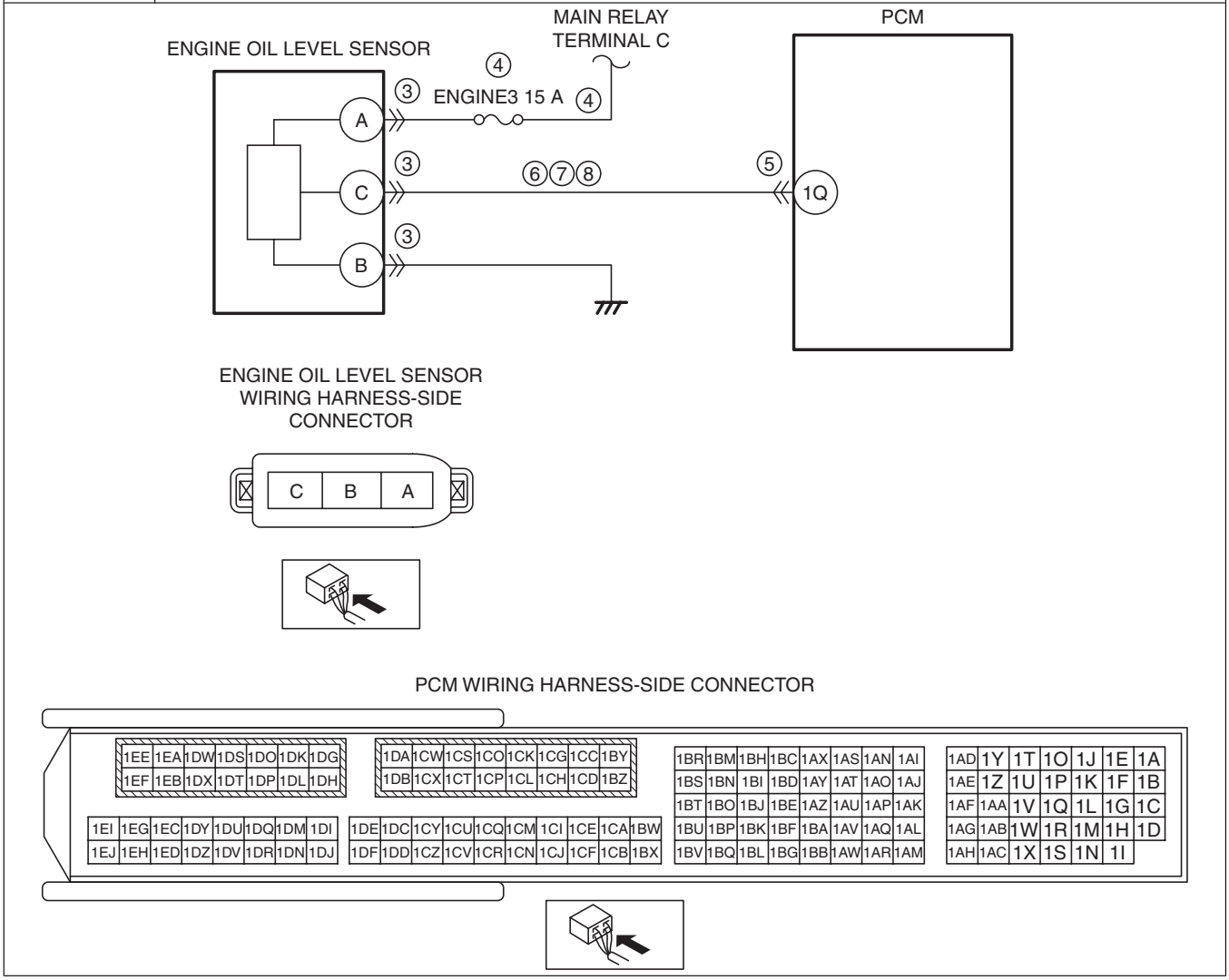
DTC P250C:00 [SKYACTIV-D 1.5]

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DTC P253F: 00	**Engine oil level sensor power supply circuit malfunction**
DETECTION CONDITION	• When the following conditions are met, a supply voltage error signal from the engine oil level sensor continues to be received for a specified time. **MONITORING CONDITIONS** — All of the following conditions is met for 1min. • Battery voltage: **10—16 V** • Communication error between oil level sensor and PCM DTC U1100:00 not detected. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Engine oil level sensor connector or terminals malfunction • Short to ground or open circuit in engine oil level sensor power supply circuit — Short to ground in wiring harness between ENGINE3 15 A fuse and engine oil level sensor terminal A — ENGINE3 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and engine oil level sensor terminal A • PCM connector or terminals malfunction • Short to ground in wiring harness between engine oil level sensor terminal C and PCM terminal 1Q • Short to power supply in wiring harness between engine oil level sensor terminal C and PCM terminal 1Q • Open circuit in wiring harness between engine oil level sensor terminal C and PCM terminal 1Q • Engine oil level sensor malfunction • PCM malfunction

DTC P253F: 00	Engine oil level sensor power supply circuit malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**PURPOSE: RECORD FREEZE FRAME DATA/ SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (misfire related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT ENGINE OIL LEVEL SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the engine oil level sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT ENGINE OIL LEVEL SENSOR POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the engine oil level sensor connector is disconnected. • Switch the ignition off. • Measure the voltage at the engine oil level sensor terminal A (wiring harness-side). • Is the voltage B+?	Yes	Go to the next step.
		No	Inspect the ENGINE3 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and engine oil level sensor terminal A. If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: • Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and engine oil level sensor terminal A. If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
6	**INSPECT ENGINE OIL LEVEL SENSOR LIN COMMUNICTION CIRCUIT FOR SHORT TO GROUND** • Verify that the engine oil level sensor and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the engine oil level sensor terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between engine oil level sensor terminal C and PCM terminal 1Q. If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
7	**INSPECT ENGINE OIL LEVEL SENSOR LIN COMMUNICTION CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the engine oil level sensor and PCM connectors are disconnected. • Switch the ignition off. • Measure the voltage between engine oil level sensor terminal C (wiring harness-side) and ground. • Is the voltage 0 V?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between engine oil level sensor terminal C and PCM terminal 1CP. If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has a short to power supply. Go to Step 9.
		No	Go to the next step.
8	**INSPECT ENGINE OIL LEVEL SENSOR LIN COMMUNICTION CIRCUIT FOR OPEN CIRCUIT** • Verify that the engine oil level sensor and PCM connectors are disconnected. • Inspect for continuity between engine oil level sensor terminal C (wiring harness-side)and PCM terminal 1Q(wiring harness-side). • Is there continuity?	Yes	Replace the engine oil level sensor, then go to the next step. (See ENGINE OIL LEVEL SENSOR REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)
		No	Refer to the wiring diagram and verify whether or not there is a common connector between engine oil level sensor terminal C and PCM terminal 1Q. If there is a common connector: If there is a common connector: • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. If there is no common connector: • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.